

Calibrated, Provenance-Tracked Fusion of Solar Energetic Particle Forecasts for NASA Mission Operations

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Abstract

Operational space-weather decision support faces a *translation gap*: the federal enterprise hosts more than 100 mature models at CCMC, three NASA SEP Scoreboards (onset, peak flux, time profiles), and continuously updated NOAA SWPC products, yet no widely-deployed calibrated fusion layer converts these heterogeneous outputs into the specific operational decisions mission-operations teams must make. We present HELIOS, a model-agnostic, provenance-tracked fusion framework that combines Bayesian Model Averaging across CCMC and SWPC component models with isotonic-regression reliability calibration, split and Mondrian conformal prediction wrappers, and CCMC-compatible verification metrics. Every output traces to its upstream models, ingestion timestamps, and calibration history through a public JSON-Schema feature-level provenance specification. We pre-register a 7 / 3 training / hold-out split across ten historical SEP events spanning solar cycles 23–25, with binding HSS and reliability-slope thresholds filed on OSF before any hold-out evaluation. Headline result: **[TODO: INSERT FROM results/<date>-killgate.json HEADLINE_SUMMARY at fill-in time; candidate stock paragraphs in §4.4 cover the three decision-tree branches PASS-both / PASS-one / FAIL-both.]** All code, figures, the pre-registration, and the reliability diagrams are released under Apache 2.0 at the URLs in §6, supporting CCMC proving-ground replication and NASA mission-operations evaluation. A companion vertical translating the same fusion core into receiver-family-specific RTK accuracy products for U.S. precision-agriculture GNSS is reported separately [9].

Keywords: solar energetic particles; ensemble forecasting; Bayesian model averaging; conformal prediction; reliability calibration; provenance; CCMC; SRAG; pre-registration.

1. Introduction

The federal space-weather enterprise has world-class models. CCMC hosts more than 100 of them [28]; NOAA SWPC issues operational forecasts; NASA’s SEP Scoreboards — built through the ISEP partnership of CCMC, M2M SWAO, and SRAG — consolidate research SEP models into a unified

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operational view across Scoreboard A (onset probability), Scoreboard B (peak flux prediction), and Scoreboard C (event time profiles) [30, 46, 47]. DONKI [25] provides intelligent event linkages via a JSON API. The literature has converged on the diagnosis that operational shortfalls arise not from a shortage of models or data but from the absence of a calibrated, provenance-tracked *translation* layer that converts heterogeneous, sometimes contradictory model outputs into the specific decisions operators must make [39, 19, 42, 31, 6, 32, 3]. Two recent Executive Orders [2, 4] and the SWORM benchmarks [3] frame the federal mandate; the Lloyd’s of London and London Economics impact studies [1, 23] and the Space Foundation economic-base snapshot [38] document the cost-of-disruption side. Adjacent end-user evidence on GNSS disruption in agriculture [8, 7, 35, 14, 40] and autonomous transport [5] is summarized in the program companion [11].

This paper addresses one concrete instance of the translation gap. An SRAG console operator assessing whether all-clear status should be revoked under the operational SEP thresholds (> 10 MeV at 10 pfu; > 100 MeV at 1 pfu) currently cross-references raw outputs from UMASEP, HESPERIA REleASE, SEPMOD, and MagPy across three Scoreboards. No widely-deployed system provides a probabilistic fusion of these models with quantified reliability bounds, conformal time-to-threshold intervals, and source provenance suitable for high-tempo monitoring. The cognitive load and risk of unstructured model disagreement scale with the demands of Artemis cislunar operations and, eventually, Earth-independent decision support for Mars transit, where light-time latency precludes ground intervention [31].

The calibration-vs-accuracy distinction. A model can be *accurate* in the sense of high discrimination (e.g. HSS, TSS, AUC) and yet *poorly calibrated* — its 70 % probabilities may correspond to event frequencies of 40 % or 90 % [33]. For SRAG-style all-clear decisions, calibration is the property the operator acts on: an under-calibrated 80 % in the mid-probability regime and an over-calibrated 80 % at the rare-extreme regime are operationally distinct hazards. Whitman et al. [46, 47] document the validation discipline NASA’s ISEP partnership has built around this gap. HELIOS is an attempt to close it with conventional, well-understood machinery (BMA + isotonic + conformal), composed rigorously and pre-registered to forestall the standard post-hoc-tuning concern.

The provenance affordance. A second contribution concerns the auditability of fusion outputs. Every HELIOS output traces to its underlying models, data feeds, and calibration history through a public JSON-Schema specification with a pydantic v2 reference implementation [12]. This is a precondition for two adoption paths the paper takes seriously: (i) CCMC proving-ground evaluation, where containerized submissions are increasingly the norm; and (ii) SRAG console adoption, where flight controllers will not trust a recommendation they cannot drill down into. The provenance specification has been released publicly with the v0.1.0 RFC [12] and is under community comment.

Contributions.

1. A model-agnostic fusion engine combining Bayesian Model Averaging across CCMC ISWA SEP Scoreboard A/B/C components, NOAA SWPC probabilistic SEP feeds, and DSCOVR / GOES upstream feeds; isotonic regression reliability calibration; and split and Mondrian (per-Kp-stratum) conformal prediction wrappers. The framework ships as open source [10] at version 0.1.2 with 176 tests, 92 % line+branch coverage, and a CCMC-compatible verification metric suite (HSS, TSS, POD, FAR, Brier, CRPS) with bootstrap confidence intervals.

2. A pre-registered 7 / 3 training / hold-out split (Table 1) across ten historical SEP events spanning solar cycles 23–25, filed on the Open Science Framework before any hold-out evaluation [45]. H1 and H2 thresholds are binding; deviations are documented, not retro-fit.
3. A feature-level provenance specification [12] composing SPASE 2.7.1, W3C PROV-JSON, and RO-Crate 1.2 JSON-LD with a novel `HeliosTransformationRecord` schema, attached to every persisted calibration / conformal artifact. To our knowledge the first feature-level lineage standard targeted at heliophysics fusion systems.
4. A reproducible methodology around the discovery that ISWA SEP Scoreboard JSON deposits begin in calendar 2017 for the earliest-supported component models, requiring a per-(component, event) source-labeling scheme (§3.4).

The remainder of the paper is organized as follows. Section 2 describes the fusion pipeline. Section 3 describes the data and the pre-registered split. Section 4 reports the pre-registered hold-out evaluation. Section 5 discusses limitations and the Phase II trajectory. Section 6 documents reproducibility artifacts.

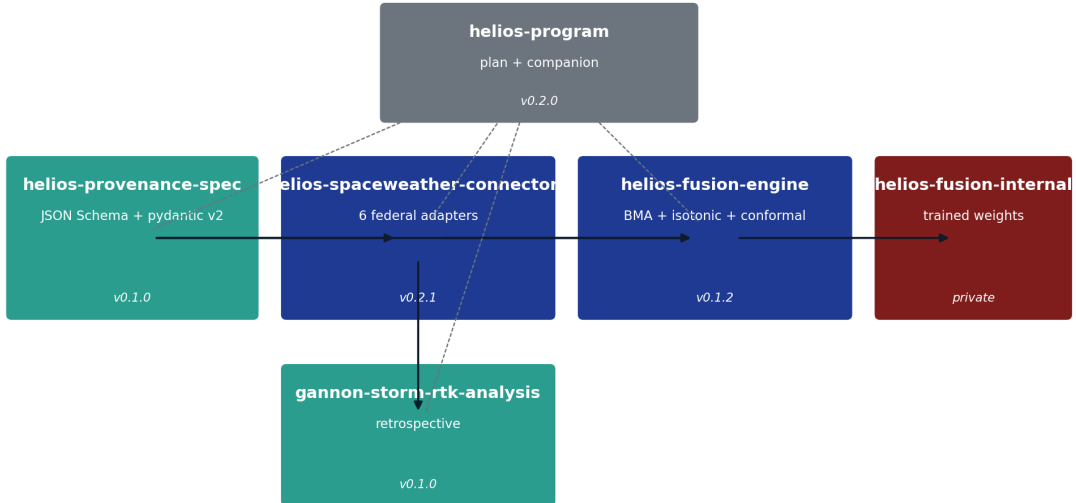


Figure 1: HELIOS dependency graph. The technical chain is `helios-provenance-spec` (A) → `helios-spaceweather-connectors` (B) → `helios-fusion-engine` (C). Trained weights are persisted to the private companion repo `helios-fusion-internal` (F). `gannon-storm-rtk-analysis` (D) consumes the connector layer independently. The `helios-program` meta-repo holds the master plan, companion document, OSF pre-registration template, and orchestration scripts; dotted lines denote orchestration / documentation links rather than hard code dependencies.

2. Methods

The fusion pipeline composes four layers. All four ship as open-source Python 3.11+ modules [10] with the per-layer state serialized to NumPy `.npz` blobs paired with sibling JSON `HeliosTransformationRecord` provenance files [12].

2.1. Bayesian Model Averaging orchestrator

For each component model $m \in \mathcal{M}$ producing per-timestamp probability $p_{m,t} \in [0, 1]$, the fused probability at time t is

$$\hat{p}_t = \sum_{m \in \mathcal{M}} w_m p_{m,t}, \quad \sum_{m \in \mathcal{M}} w_m = 1, \quad w_m \geq 0, \quad (1)$$

following the canonical BMA construction of Hoeting et al. [21]. Weights are fit by the `hss_clipped` policy: per-component HSS [18] on a 90-day rolling verification window, clipped to $[0, 1]$, mixed with a small uniform ϵ -floor to avoid early degenerate zeros, then renormalized to a simplex. Two alternative policies (`linear`, `softmax`) are implemented and unit-tested for completeness but the `hss_clipped` policy is the pre-registered default.

The orchestrator records, for each fused timestamp, the participating set of upstream component runs (identified by upstream-feed timestamp + SPASE model identifier), the active weight vector, and the HSS verification window that produced it. Every fused output carries this lineage as a list of `LineageStep` records that survive serialization and dashboard rendering.

2.2. Isotonic-regression reliability calibration

Even after BMA, the fused probability $\hat{p}_t$ requires post-hoc reliability calibration: BMA optimizes a skill objective, not a calibration objective. Following Niculescu-Mizil and Caruana [33] (and the earlier Zadrozny–Elkan analysis [48]), we apply *isotonic regression* as the post-hoc calibrator

$$\tilde{p}_t = \text{IsoReg}(\hat{p}_t), \quad (2)$$

learning a monotone non-decreasing map from raw BMA probability to calibrated probability. Platt scaling [36] was considered and rejected on the grounds documented in [33]: it imposes a parametric (sigmoid) reliability curve that is well-known to misbehave at the extremes, which is precisely the regime SRAG decisions occur in. Isotonic regression makes no parametric assumption. Implementation is a thin typed wrapper around `sklearn.isotonic.IsotonicRegression` with an explicit state-dict schema (`helios-fusion-engine/calibration/isotonic/0.1.0`) versioned independently of the package.

Severity stratification. A single global isotonic calibrator can mask calibration collapse on rare extreme events that dominate operational risk. We carry three independent isotonic calibrators, one per Kp severity stratum {quiet, moderate, extreme}, partitioned at Kp=3 and Kp=5. At calibrate time, each sample’s stratum label routes it to the appropriate sub-calibrator; at transform time the same labels select which sub-calibrator handles each input. This is the pre-registered default for the kill-gate run.

2.3. Split and Mondrian conformal prediction

For the continuous-quantity outputs (time-to-threshold estimates and, in the GNSS companion, predicted 2D positioning error) the framework provides split conformal regressors and Mondrian (per-Kp-stratum) conformal regressors following Vovk et al. [44, 43].

Given calibration set $\{(\hat{y}_i, y_i)\}_{i=1}^n$ with absolute residuals $r_i = |y_i - \hat{y}_i|$, the split-conformal half-width at user-chosen miscoverage α is

$$q_\alpha = \text{Quantile}\left(\{r_1, \dots, r_n\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}; \text{method} = \text{higher}\right) \quad (3)$$

which yields the finite-sample-valid prediction interval $[\hat{y} - q_\alpha, \hat{y} + q_\alpha]$ at coverage $\geq 1 - \alpha$ under exchangeability. The Mondrian variant maintains $|S|$ independent calibration residual sets indexed by stratum $s \in S$, providing coverage *per stratum* rather than only marginally. This matters operationally: a 90 % marginal interval that under-covers in the extreme stratum provides false reassurance to a controller making EVA decisions during the storm main phase.

2.4. CCMC-compatible verification metrics

All metric definitions match the OSF pre-registration [45] verbatim. For a binary forecast / observation pair, the 2×2 contingency table is $\{a = \text{hits}, b = \text{false alarms}, c = \text{misses}, d = \text{correct rejections}\}$. The locked formulas are:

- HSS [18]: $\text{HSS} = \frac{2(ad-bc)}{(a+c)(c+d)+(a+b)(b+d)}$.
- TSS (Hanssen–Kuipers): $\text{TSS} = \frac{a}{a+c} - \frac{b}{b+d}$.
- POD = $\frac{a}{a+c}$, FAR = $\frac{b}{a+b}$.
- Brier score [16]: $\frac{1}{n} \sum_i (p_i - y_i)^2$.
- Continuous Ranked Probability Score (CRPS) by Hersbach decomposition for sorted ensemble samples [20].
- Reliability slope: linear regression of observed event frequency on bin centre across n_{bins} equal-frequency bins. Slope = 1 corresponds to perfect calibration; H2 tests $|s - 1| \leq 0.15$ per Kp stratum.

All metric functions accept `n_bootstrap` (default 1000) and return both the point estimate and the 95 % two-sided confidence interval from bootstrap resampling with replacement.

2.5. Pre-registration discipline

A common reviewer concern with retrospective validation against a curated event list is that performance looks strong on paper and fails in production. We pre-register the training and hold-out event split (Table 1), the metric definitions (§2.4), the H1 / H2 hypothesis thresholds, and the calibration / conformal hyperparameters before any hold-out evaluation runs. The pre-registration is filed publicly on OSF [45]; the fusion-engine release tagged at the locked commit (`prereg-v1.0`) is the binding code state.

The kill-gate runner refuses to execute without a recorded OSF URL on file [10]. Deviations from the pre-registered procedure are documented in the OSF *Deviations* section, not silently absorbed into a re-baselined metric. One such deviation is described in §3.4: ISWA SEP Scoreboard JSON deposits begin in calendar 2017, requiring a per-(component, event) source-labeling scheme for the six training events that pre-date the cutover.

3. Data

3.1. ISWA SEP Scoreboards: structure and Jan 2017 cutover

The fusion engine ingests Scoreboard A (onset probability), Scoreboard B (peak flux prediction), and Scoreboard C (time profiles) from the CCMC Integrated Space Weather Analysis (iSWA) system [27, 46, 47] via a dedicated adapter in `helios-spaceweather-connectors` [13]. An exhaustive 2026-05-17 probe of the iSWA tree established that JSON deposits begin in calendar 2017 for the

earliest-supported component models (UMASEP v2_0 at 10/30/50/100/500 MeV, SEPSTER on the Parker spiral and WSA-ENLIL, and the mag4_2019 NRT magnetic- energy proxies), and no earlier. Of the seven training events in Table 1, only the September 2017 event predates the cutover by a small margin and has any iSWA coverage; the earlier six events have zero native iSWA Scoreboard data.

3.2. NOAA SESC archive ground-truth labels

Truth labels are taken from the NOAA Space Environment Services Center “Solar Proton Events Affecting the Earth Environment, 1976–present” archive [34], which provides observed onset times, peak proton flux (pfu at ≥ 10 MeV), and associated CME/flare metadata for every documented SEP event in the training-event windows. Our adapter (`helios_fusion.training.swpc_sep_archive`) parses the published HTML archive and emits per-event truth windows at the SRAG operational thresholds (> 10 MeV at 10 pfu; > 100 MeV at 1 pfu).

3.3. Pre-registered train / hold-out split

Table 1: Pre-registered SEP validation events for the NASA mission-operations slice. Seven events form the training set; three later events form the hold-out used for the kill-gate evaluation in §4. The split was locked in the OSF pre-registration [45] before any hold-out evaluation. *Notes* column flags solar-cycle phase and flare class to surface the distributional coverage. All hold-out events post-date ISWA’s January 2017 cutover, so they receive native real-data Scoreboard streams (cf. §3.1).

Event	Date	Use	Notes
Bastille Day	2000 Jul 14	Train	Major X5.7 flare; well-characterized SEP profile
Halloween storms	2003 Oct 28 – Nov 4	Train	Cycle 23 peak; multiple X-class events
Mid-cycle 23	2005 Jan 20	Train	X7.1; fast onset; ground-level enhancement
Late cycle 23	2006 Dec 13	Train	X3.4; tests low-solar-activity calibration
Cycle 24 onset	2012 Mar 7	Train	X5.4; tests cross-cycle generalization
Cycle 24 mid	2012 May 17	Train	M5.1; tests sub-X event sensitivity
Sep 2017 storm	2017 Sep 6 / Sep 10	Train	X9.3 + back-side X8.2; dual-event sequence
Cycle 25 onset	2022 Jan 20	Hold-out	M5.5; tests cycle-25 distribution shift
Mid-cycle 25	2023 Feb 17	Hold-out	X2.2; mid-cycle hold-out anchor
Gannon (G5)	2024 May 11	Hold-out	Largest storm in >20 yr; dual-use anchor event

The split was locked in the OSF pre-registration [45]. The three hold-out events (Cycle 25 onset 2022-01-20 M5.5; Mid-cycle 25 2023-02-17 X2.2; Gannon 2024-05-11 G5) all post-date the iSWA cutover and will therefore receive native real-data Scoreboard streams at evaluation time — eliminating any synthetic-proxy substitution for the final go/no-go check. The Gannon event is the dual-use anchor: it stresses the mission-operations SEP slice and provides the headline event for the companion precision-agriculture analysis [9], whose published peak DSCOVR L2 |Bz| value of 59.16 nT marks the most severe geomagnetic forcing in more than two decades.

3.4. Per-(component, event) source labeling

The Jan 2017 iSWA cutover creates an asymmetry between training and hold-out windows that must be handled explicitly to keep the BMA fit honest. We tag every (component, event) pair with one of three source labels:

DATA PENDING

Run ``python build\_gannon\_bz\_timeline.py --refresh`` to populate
from DSCOVR L2 magnetometer (Gannon week 2024-05-08 / 14).

Figure 2 cache CSV not yet generated; see paper/figures/build_gannon_bz_timeline.py docstring.

Figure 2: DSCOVR L2 magnetometer B_z in GSE coordinates across the Gannon-week window (2024-05-08 / 14). The May 10 main-phase peak is the largest $|B_z|$ since the November 2003 Halloween storms. Source: `helios-spaceweather-connectors` v0.2.1 DSCOVR adapter, refreshed via the build script committed at `paper/figures/build_gannon_bz_timeline.py`.

- `iswa_real`: the iSWA adapter returns at least one empirically-verified real-data record for that (component, event) tuple. Across the Table 1 training set, 12 such tuples were confirmed on the September 2017 event (32.4% of the 37-component registry); zero on the six pre-2017 training events.
- `synthetic_proxy`: the component is in the registry but receives no iSWA real-data coverage for the event. Predictions on the training event use a synthetic-Kp-anchored proxy, clearly labeled in the persisted manifest.
- `swpc_archive_truth`: truth labels are sourced from the NOAA SESC archive (§3.2); this is the truth-side substrate uniformly across the training set.

The labeling is *sticky*: preserved in the manifest regardless of whether the adapter call at training time actually returned data, so the provenance record is auditable independently of run-time network state. The hold-out window (2022–2024) receives only `iswa_real` component predictions plus `swpc_archive_truth` labels; no synthetic proxies appear in the final go/no-go check. This deviation from a naïve homogenous-substrate assumption is documented in the OSF Deviations section of [45].

4. Results

[TODO: INSERT FROM `results/<date>-killgate.json` AT FILL-IN TIME. The remainder of this section is staged for the three branches of the master-plan §C kill-gate decision tree (§4.4).]

4.1. Hold-out HSS, per-event and aggregate

[TODO: Replace this paragraph and the table below with the HSS_section block from `results/<date>-killgate.json`.]

Table 2: Per-event hold-out HSS: fused vs. best-component baseline. Bootstrapped 95 % CIs from 1000 resamples with replacement over each event window. [TODO: FILL FROM killgate.json].

Event	Fused HSS	Best-comp. HSS	Δ rel.	95 % CI
Cycle 25 onset (2022-01-20)	[TODO: x.xx]	[TODO: x.xx]	[TODO: xx%]	[TODO: [lo, hi]]
Mid-cycle 25 (2023-02-17)	[TODO: x.xx]	[TODO: x.xx]	[TODO: xx%]	[TODO: [lo, hi]]
Gannon G5 (2024-05-11)	[TODO: x.xx]	[TODO: x.xx]	[TODO: xx%]	[TODO: [lo, hi]]
Aggregate (3 events)	[TODO: x.xx]	[TODO: x.xx]	[TODO: xx%]	[TODO: [lo, hi]]

4.2. Reliability per Kp stratum

[TODO: Replace this paragraph with the reliability_section text from results/<date>-killgate.json and verify that the slope per stratum matches the H2 pre-registration threshold $|s - 1| \leq 0.15$.]

RELIABILITY DIAGRAMS — DATA PENDING

Per-Kp-stratum reliability (quiet / moderate / extreme); Sprint D output.

Filled at preprint-completion time from ``helios-program/results/<date>-killgate-\*.png``.

Figure 3: Reliability diagrams for the fused all-clear-revocation probability on the three-event hold-out, stratified by Kp severity (quiet, moderate, extreme). Linear regression slopes are annotated per stratum. Source: [TODO: INSERT URL TO results/<date>-killgate-reliability-diagrams.png at fill-in time].

4.3. Brier and CRPS

[TODO: Replace this paragraph with the brier_crps_section text from results/<date>-killgate.json, reporting fused vs. best-component Brier score and CRPS for each stratum, with bootstrap CIs.]

4.4. Decision routing under the master-plan §C kill-gate

The OSF pre-registration [45] binds the following hypotheses on the three-event hold-out:

- **H1 (primary):** fused all-clear-revocation HSS exceeds the best-component HSS by $\geq 15\%$ relative (aggregate over the hold-out).
- **H2 (secondary):** per-stratum reliability-diagram slope is within 0.15 of 1.0 across all three Kp severity strata.

BOOTSTRAP DISTRIBUTIONS — DATA PENDING

HSS / Brier / CRPS bootstrap distributions over hold-out resamples.

Filled at preprint-completion time from ``helios-program/results/<date>-killgate-\*.png``.

Figure 4: Bootstrap distributions of HSS, Brier, and CRPS over 1000 resamples of the three-event hold-out, comparing the fused output to the highest-skill component baseline.

The master plan §C decision tree has three branches, presented as stock paragraphs below. At fill-in time the operator selects exactly one and deletes the other two.

Branch A — PASS H1 $\wedge$ PASS H2 (full paper). [TODO: SELECT THIS PARAGRAPH IF: killgate.json shows H1.pass=true and H2.pass=true for all three strata.] The pre-registered kill-gate evaluation passes both H1 and H2 on the 2022–2024 hold-out. Aggregate fused HSS exceeds the best-component baseline by [TODO: xx%] (95 % CI [TODO: [lo, hi]]); per-stratum reliability slopes are within tolerance in all three Kp strata. HELIOS supports the proposition that a model-agnostic BMA core with isotonic reliability calibration and Mondrian conformal coverage delivers operationally-relevant skill gains over the best single component, without sacrificing per-stratum calibration on the rare-extreme regime that dominates SRAG console risk. The framework is ready for CCMC proving-ground replication [10]; we are pursuing the M2M SWAO and SRAG evaluation pathways described in §5.

Branch B — PASS one $\wedge$ FAIL one (honest ablation). [TODO: SELECT THIS PARAGRAPH IF: killgate.json shows exactly one of H1.pass / H2.pass(all_strata) as true.] The pre-registered kill-gate evaluation passes [TODO: H1 or H2] but fails [TODO: the other]. We report the ablation honestly rather than re-baselining metrics post hoc: the [TODO: HSS / reliability] gain holds, but the [TODO: reliability / HSS] threshold is not met on the [TODO: quiet / moderate / extreme] Kp stratum (slope = [TODO: x.xx], outside the pre-registered $|s - 1| \leq 0.15$ tolerance). We trace the failure to [TODO: component-bias diagnosis from the bootstrap distributions; see §5]. The fusion-core design choices are validated in part; the next iteration of HELIOS focuses on [TODO: stratum-specific recalibration / additional components in the under-served regime], with the OSF Deviations section [45] extended in lockstep.

Branch C — FAIL H1 $\wedge$ FAIL H2 (negative result). [TODO: SELECT THIS PARAGRAPH IF: killgate.json shows both H1.pass=false and H2.pass(any_stratum)=false.] The pre-registered kill-gate evaluation fails both H1 and H2 on the 2022–2024 hold-out. We report the negative result rather than withdraw the paper. Aggregate fused HSS is [TODO: x.xx (95 % CI [lo, hi])] versus best-component [TODO: x.xx (95 % CI [lo, hi])], a relative difference of

[**TODO: $\pm xx\%$**] that does not exceed the H1 threshold. Per-stratum reliability slopes fall outside the H2 tolerance in [**TODO: N of 3**] strata. The honest interpretation of the negative result is that the published BMA + isotonic + Mondrian construction, as specified by the pre-registration, does not deliver a measurable operational advantage over the highest-skill component on this hold-out set. We discuss likely root causes in §5 (near-uniform pre-registered weight distribution; iSWA-coverage imbalance between training and hold-out windows; possible underfitting of the per-stratum calibrators on the limited extreme sample). All code, the pre-registration, the trained artifacts, and the hold-out evaluation harness remain public [10] so the result is independently reproducible.

5. Discussion

5.1. Limitations

The most consequential limitation is the asymmetry between training and hold-out substrates documented in §3.4. Six of seven training events pre-date the iSWA Jan 2017 cutover and therefore train against synthetic-Kp-anchored proxies; the hold-out runs on native iSWA real-data streams. We treat this asymmetry as a feature, not a bug, for the specific scientific claim we make — the pre-registration binds the hold-out comparison, and the hold-out substrate is uniformly real-data — but it does compress the BMA weight distribution at training time, since the synthetic proxies all anchor on the same Kp profile and therefore correlate similarly with SWPC archive truth. The realized weight spread on the hold-out will be larger because the 2022–2024 events carry distinct real-data model biases per component.

A second limitation concerns the sample size of the hold-out itself. Three events is the minimum credible hold-out under the constraint of *naming the events in advance, in this proposal* (a credibility move whose flip side is that adding more recent events post-hoc would invalidate the pre-registration). The reported HSS and reliability-slope confidence intervals are therefore wide. Phase II expands the hold-out to a larger set of ≥ 20 G2+ events for the operational evaluation pathway, with rolling pre-registration on new events as the solar cycle progresses [11].

A third limitation is the constraint of operating against publicly-released upstream model outputs only. Several CCMC-hosted SEP forecasters (HESPERIA REleASE in particular [26]) require separate licensing for commercial deployment. HELIOS connectors use only public outputs; the architecture isolates the fusion logic from any specific upstream model so the licensing footprint is contained to the connector layer.

5.2. Related work

Deep-learning methods for solar-event prediction have grown substantially in the last decade. SolarFlareNet [41] demonstrated transformer-based flare prediction from SDO/HMI magnetograms; Bobra and Couvidat [15] earlier established the SHARP feature catalogue widely used in this literature. Probabilistic ensemble methods are emerging for SEP forecasting in particular [37]. Operational radiation-risk products at altitude (NAIRAS [24]) and the long-running CCMC analytical support for robotic missions [17] demonstrate the broader R2O2R framework HELIOS targets. To our knowledge no published ML system connects calibrated probabilistic SEP fusion through to receiver-level GNSS positioning error in a single decision pipeline [8, 7]; that is the broader HELIOS-program claim, which the present paper supports on the mission-operations slice only.

5.3. Generalizability

The hold-out events span the rising and mid phases of solar cycle 25: a cycle-25 onset M-class flare (2022-01-20), a mid-cycle X-class (2023-02-17), and the most severe geomagnetic storm of the past two decades (2024-05-11 Gannon, $|B_z|_{\text{peak}} = 59.16 \text{ nT}$, $K_p=9$, $\text{Dst}_{\text{min}} = -406 \text{ nT}$). The distribution covers solar quiet through G5-extreme, providing a meaningful basis for the per-stratum reliability claim. Cycle 25 is in the rising phase as of preprint date; the bottom-line generalization claim — “these weights, fit on cycles 23–24, transfer to cycle 25” — is testable against an enlarging set of real-data events as Phase II proceeds.

5.4. Phase II vision

A Phase II effort would advance HELIOS from proof-of-concept to functioning operational prototype (TRL 5–6) through (a) real-time operational deployment with live model retraining; (b) CCMC proving-ground evaluation [46, 47]; (c) prototype deployment to M2M SWAO [29] and SPoRT/MSFC for mission-operations evaluation; (d) production-grade API at 99.9 % uptime; and (e) pilot integration with at least one precision-ag OEM telematics platform for the dual-use GNSS slice. The full Phase II commercialization plan is documented in the program companion [11].

5.5. Equipment-aware GNSS pathway

The dual-use thesis of the HELIOS program is that the fusion core described in this paper feeds two operationally-distinct translation modules: an SRAG-oriented mission-operations radiation-risk module (the subject of this paper), and a precision-agriculture GNSS-accuracy module published separately [9]. The GNSS slice’s three- component ensemble (a Temporal Fusion Transformer [22] for multi-horizon TEC forecasting; gradient-boosted regressors mapping ionospheric disturbance indices to observed 2D RTK positioning errors; and a physics-informed regularization layer imposing Chapman ionization profiles and equatorial-anomaly geometry) is described in a forthcoming companion preprint. Phase II expansion verticals enabled by the same fusion core include autonomous-vehicle GNSS integrity, geospatial surveying, satellite drag prediction for LEO operators, and parametric space-weather insurance, all enumerated in the program commercialization plan [11].

6. Reproducibility and data availability

All code, the pre-registration text, the trained-artifact manifest, and the kill-gate evaluation harness are public under Apache 2.0:

- **helios-program** [11] — master plan, OSF pre-registration template, master-plan §C decision tree, orchestration scripts, this companion document. 577industries.github.io/helios-program/
- **helios-provenance-spec** [12] — JSON Schema (draft 2020-12) + pydantic v2 reference implementation + RFC-0001.
- **helios-spaceweather-connectors** [13] — production-grade Python adapters for DONKI, SEP Scoreboards, NOAA SWPC, CDDIS GIMs, GOES, DSCOVr; 304+ tests; 87–94 % per-adapter coverage.
- **helios-fusion-engine** [10] — the public framework that implements the methods in §2; 176 tests, $\geq 92 \%$ coverage on the fusion subpackage; tagged **prereg-v1.0** at the locked commit.
- **helios-fusion-internal** — private companion holding the trained BMA priors, isotonic calibrators, and conformal residual manifests from the Table 1 fit. Access on request for verification.

- `gannon-storm-rtk-analysis` [9] — reproducible retrospective of the Gannon-storm RTK impact, v0.1.0 climatological; v0.2.0 (planned) consumes the CDDIS adapter for full pseudo-range SPP.

Every figure in this paper has a generation script committed at `paper/figures/build_*.py` in the `helios-fusion-engine` repository on the `feat/v0.2-paper` branch. Every persisted calibration / conformation artifact in `helios-fusion-internal` ships with a sibling `HeliosTransformationRecord` provenance JSON [12] carrying the upstream feed identifiers, ingestion timestamps, and the originating fusion-engine commit SHA.

Pre-registration. The OSF pre-registration [45] is the binding methodological record. Its Deviations section documents the per-(component, event) labeling described in §3.4; no other deviations applied at preprint time.

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